TrueSkate v1 - every net, in plain English

Identical labels are one electrical net. Shared copper remains continuous with any axis unpopulated. The two shared kill buffers and their bias/bypass parts remain fitted. Plug-in modules perform motor control.

NET / FAMILY	CONNECTION AND PURPOSE
12V_IN	J5 positive to F1 input, before the fuse. 3+3 top-layer input route; correct polarity required.
12V_PREPOL	F1 output to Q1 drain before reverse-polarity protection; unprotected positive input.
POL_GATE	Q1 gate, R2 pull-down to GND and D1 zener anode; D1 cathode joins the source.
12V_FUSED	Q1 protected source output to all Mini VIN contacts and local capacitors. 3+3 top-layer backbones and feeds; 1 oz copper.
3V3	Teensy regulator to encoders, CS pull-ups, kill buffers and /OE pull-up. USB powers the Teensy.
GND	Stitched top/bottom GND fills join supply negative, logic, Minis and capacitor returns. No isolation.
SPI_SCK	Teensy 13 to all encoders. SH contact assignment is provisional; see README verification table.
SPI_MOSI	Teensy 11 to all encoders. Host-to-encoder data.
SPI_MISO	Encoders to Teensy 12. Only the selected encoder may drive this shared line.
A<n>_PWM<p>	Three logic PWM instructions per axis from Teensy to Mini. Existing pin map retained.
A<n>_EN_CMD	Teensy EN command to a buffer input; 10k input pull-down holds an undriven command low.
EN_DISABLE	J6 NC loop shorts this net to GND to permit commands. Open loop lets R1 pull it high and disable all buffers.
A<n>_EN	Buffered enable to Mini pin9. Open kill loop makes buffer high impedance; local 10k pulls EN low.
A<n>_CS	Individual active-low encoder select with 10k pull-up. Fit with its encoder; deselect all before SPI.
A<n>_PH<p>	Mini P3/P2/P1 to male motor header1/2/3. Never move phase cables with 12V live.

TEENSY PIN NUMBERS (not socket contact numbers)

AXIS / MOVEMENT	PWM1 / PWM2 / PWM3	EN	CSn
A1 / index curl 1	4 / 5 / 6	16	26
A2 / index curl 2	2 / 22 / 23	17	27
A3 / index splay	7 / 36 / 29	20	32
A4 / middle curl 1	24 / 0 / 25	21	38
A5 / middle curl 2	3 / 33 / 37	30	39
A6 / middle splay	8 / 9 / 28	31	40

SH pin-to-net order is UNVERIFIED: see README and check the actual AS5048A board with a multimeter. Opening kill disables EN but does not disconnect 12V; reclosing permits the current command without a latch. Human review, mechanical fit, startup and noise tests remain required before manufacture.